

Courant Algebroid Relations and T-duality

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Outline

- 1 T-duality
- 2 Courant algebroids
- 3 CA Relations
- 4 T-duality again

Basic example

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A string with momentum w and winding number p , moving on a circle of radius $1/R$ actually describes the same physics.

So

$$p \rightarrow w, \quad w \rightarrow p, \quad R \rightarrow 1/R$$

sees a T-duality.

T-duality as an isometry

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This comes with a natural metric

$$\langle X + \xi, Y + \eta \rangle = \eta(X) + \xi(Y).$$

T-duality can be seen as an isometry of this metric, i.e. a map φ with

$$\langle \varphi \cdot, \varphi \cdot \rangle = \langle \cdot, \cdot \rangle.$$

Basic example revisited

Returning to our basic example, we have two circle bundles over the same base, and a map between their respective generalised tangent bundles.

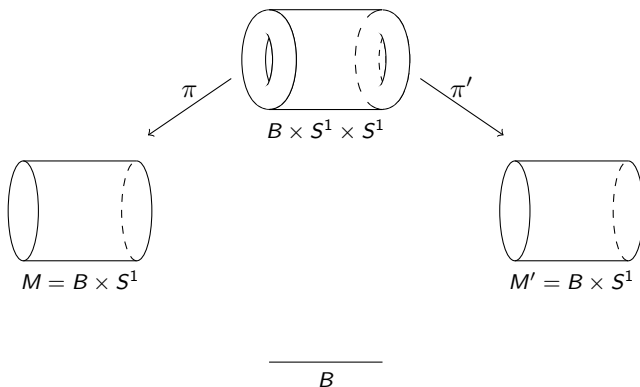
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We can formalise this map by introducing the correspondence space, over the same base, as an $S^1 \times S^1$ -bundle.

The idea is to lift a generalised vector to the correspondence space in such a way that it projects down to the T-dual space.

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Courant algebroids

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For the generalised tangent bundle $TM \oplus T^*M$, involutivity of a subbundle is measured with respect to the *Courant bracket*

$$[X + \xi, Y + \eta] = [X, Y] + \mathcal{L}_X \eta - (d\xi)(Y).$$

In general, a *Courant algebroid* is a vector bundle $E \rightarrow M$ equipped with a non-degenerate metric, a bracket, and an *anchor* $\pi : E \rightarrow TM$, satisfying various compatibility conditions (e.g. Leibniz).

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It is *exact* if it fits in the exact sequence

$$0 \rightarrow T^*M \rightarrow E \rightarrow TM \rightarrow 0,$$

and hence isomorphic to $TM \oplus T^*M$.

Courant algebroids

A T-duality is an isomorphism of Courant algebroids, i.e. a φ such that

$$\langle \varphi \cdot, \varphi \cdot \rangle = \langle \cdot, \cdot \rangle, \quad \text{and} \quad [\varphi \cdot, \varphi \cdot] = \varphi[\cdot, \cdot].$$

This allows to carry information about involutive structures between the Courant algebroids.

Reduction

Suppose we have a principal G -bundle M over a base B (so that $B = M/G$), and an exact Courant algebroid E on M . Can we form a Courant algebroid on B ?

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There is an action of the Lie algebra $\mathfrak{g}$ on TM , and we can extend this to an action on E , so there is the bundle E/G on B , but not necessarily exact.

Instead, take

$$\underline{E} = \frac{K^\perp}{K \cap K^\perp} / G,$$

where K is the image of the action $\mathfrak{g} \rightarrow \Gamma(E)$.

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so that it is a CA isomorphism.

Compositions

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E.g. $\text{gr}(\Phi') \circ \text{gr}(\Phi) = \text{gr}(\Phi' \circ \Phi)$.

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Involutive structures

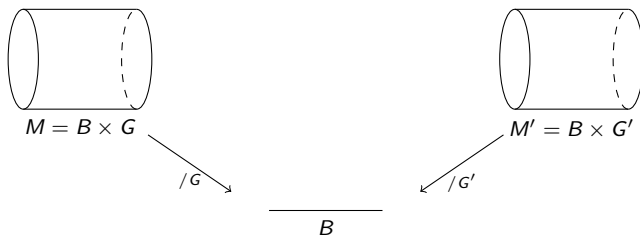
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The product $\{0\} \times L$ defines a relation from $\{0\}$ to E , so $R_*(L)$ is determined uniquely by

$$\{0\} \times R_*(L) = R \circ (\{0\} \times L).$$

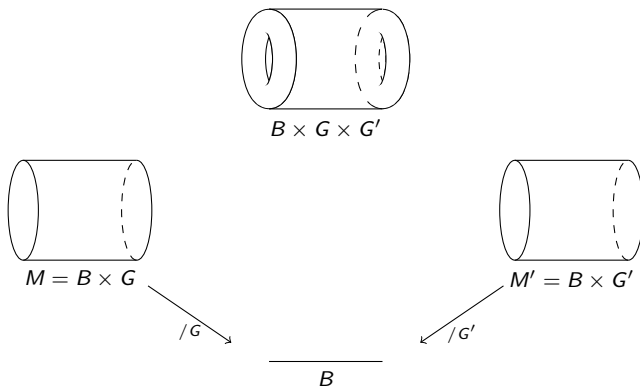
T-duality of G -bundles

Suppose we have a G -bundle, a G' -bundle, and a $G \times G'$ -bundle over the same base.



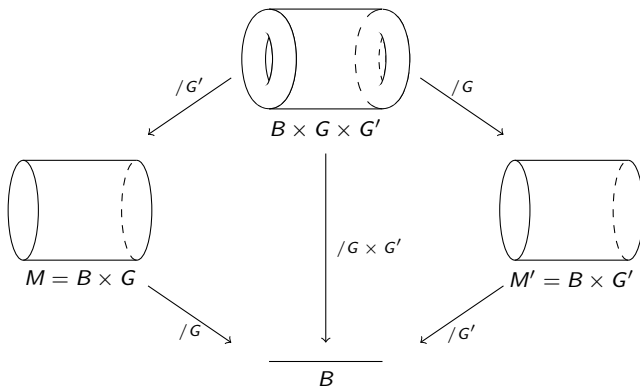
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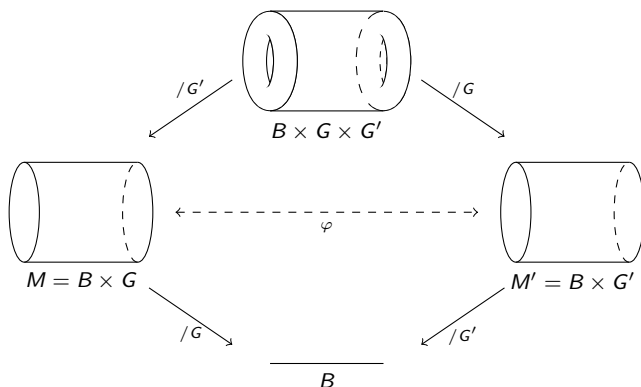
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On the CA level, we can package the reduction procedure into a CA relation.

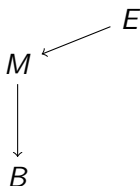
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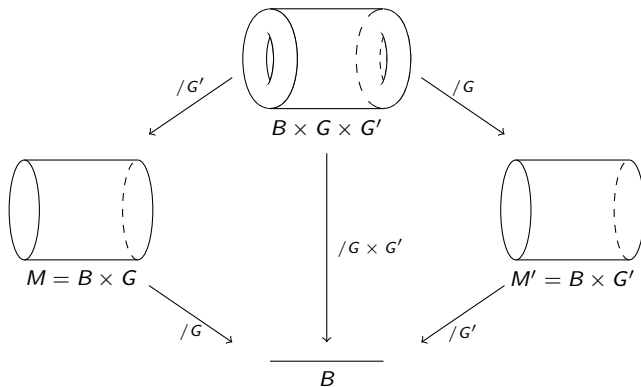
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Since P is not a vector bundle map on E , we look at the relation

$$Q(G) = \{(e, P(e)) : e \in K^\perp\} \subset E \times \underline{E}.$$

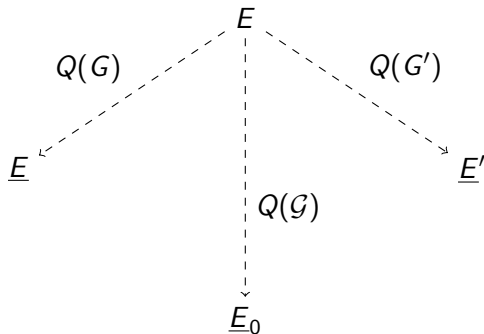
Base picture

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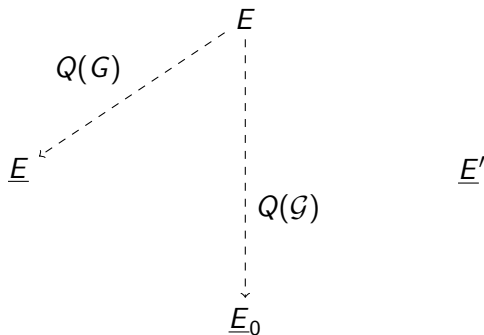
On the CA level

We have, the relations $Q(G)$, $Q(G')$ and $Q(G \times G' = \mathcal{G})$.



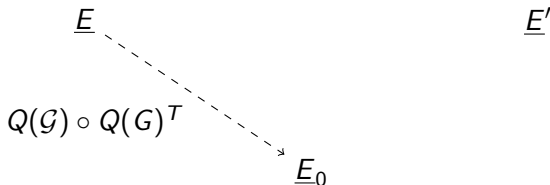
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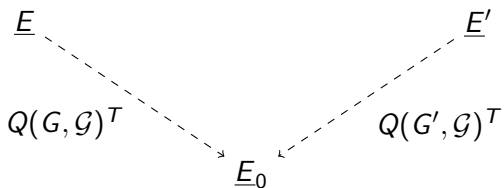
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$$\underline{E} \overset{Q(G', \mathcal{G})^T \circ Q(G, \mathcal{G})}{\dashrightarrow} \underline{E}'$$

$$\underline{E}_0$$

Involutive Structures

It turns out that this relation gives a bijection between the involutive subbundles of $\underline{E}$ and those of $\underline{E}'$.

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$Q(G', \mathcal{G})^T \circ Q(G, \mathcal{G})$ is called the *Poisson-Lie T-duality*.